HYPOXIA

Dr Shahneela Siraj
Associate Professor
DIKIOHS, DUHS

LEARNING OBJECTIVES

At the end of this lecture the students should be able to:

- Define hypoxia
- Describe the hypoxic injury to cell.
- Explain the types of hypoxia.
- Elucidate the causes of hypoxia.
- Describe cyanosis.
- Discuss the effects of hypoxia.

HYPOXIA

Hypoxia is defined as a decrease in O2 delivery to, or utilization by, the tissues.

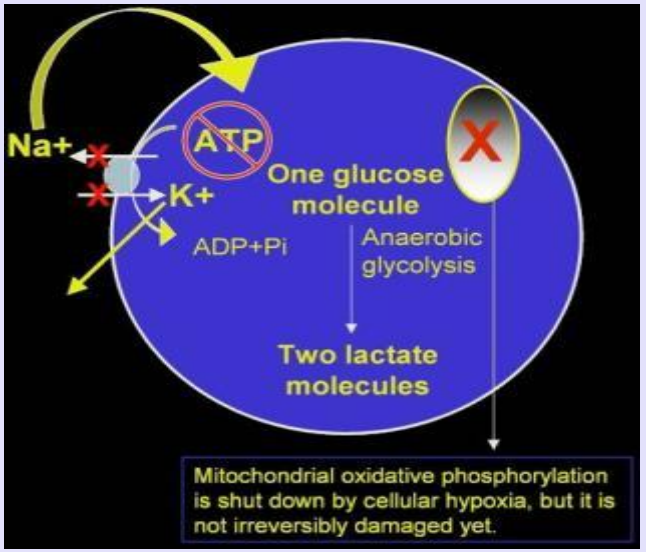
Hypoxemia is defined as a decrease in arterial PO2.

Hypoxemia is one cause of tissue hypoxia, although it is not the only cause.

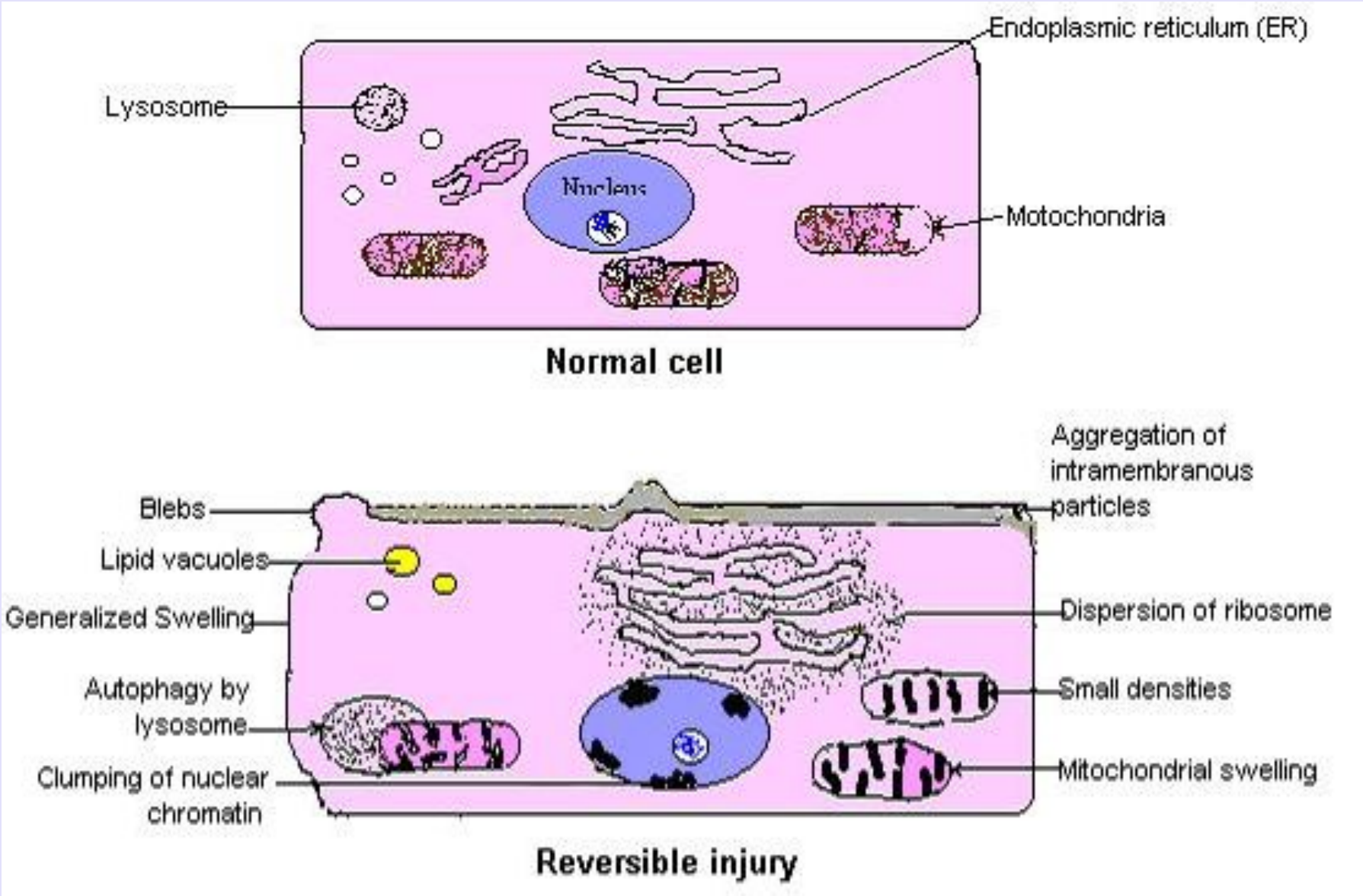
HYPOXIC INJURY TO CELLS

- **REVERSIBLE CHANGES**:
  - Impaired aerobic respiration (mitochondria)
  - Decreased ATP (energy)
  - Anerobic glycolysis
  - Glycogen depletion
  - Accumulation of lactic acid
  - (intracellular acidosis)

  with associated nuclear chromatin clumping.



REVERSIBLE CELL INJURY



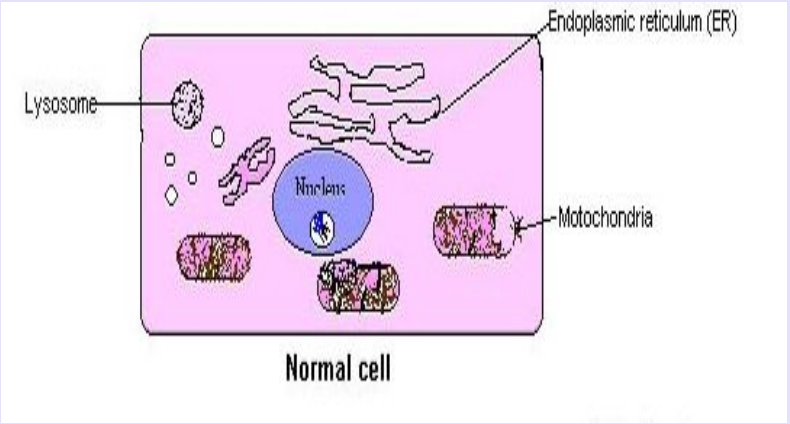
HYPOXIC INJURY TO CELLS

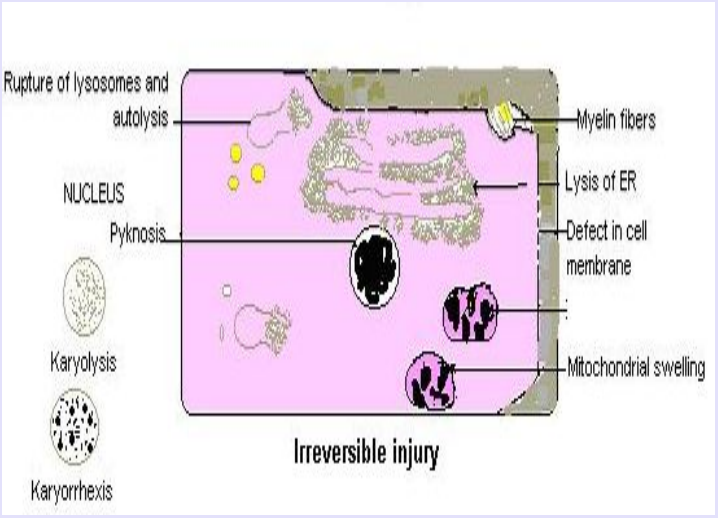
IRREVERSIBLE changes:

- **Vacuolization of mitochondria**
- **Swelling of lysosomes**
- **Damage to plasma membranes**
  - **Loss of phospholipids (dec. synthesis and inc. degradation).**
  - **Cytoskeletal alterations (damage to cytoskeletal-membrane connections,**

    effects of cell swelling,

    activation of proteases)
  - **Effects of free radicals (toxic oxygen radicals produced by PMN's)**





HYPOXIC INJURY TO CELLS

IRREVERSIBLE INJURY

- Lipid breakdown products (free fatty acids and other with a detergent effect on cell membrane)
- Influx of Ca++ into the cell and mitochondria with inhibition of cellular enzymes
- Denaturation of proteins and coagulation of cells (coagulative necrosis)
- Cell components are degraded by inflammatory processes, with associated further enzyme leakages and release of inflammatory mediators
- Final breakdown product of dead cells include free fatty acids which attract Ca++ with formation of soaps

TYPES OF HYPOXIA

There are 4 types of hypoxia:

- Hypoxic Hypoxia
- Anemic Hypoxia
- Stagnant / Ischemic Hypoxia
- Histotoxic Hypoxia

HYPOXIC HYPOXIA

- **Hypoxic** hypoxia means decreased oxygen content in blood. It is also called arterial hypoxia.
- Hypoxic hypoxia is characterized by reduced oxygen
- tension in arterial blood.
- In other types of Hypoxia, PO2 is normal.

CAUSES

- Hypoxic hypoxia is caused by four factors.

  i. Low oxygen tension in inspired (atmospheric) air, which does not provide enough oxygen.

  ii. Respiratory disorders associated with decreased pulmonary ventilation, which does not allow intake of enough oxygen.

  iii. Respiratory disorders associated with inadequate oxygenation in lungs, which does not allow diffusion of enough oxygen.

  iv Cardiac disorders, in which enough blood is not pumped to transport oxygen.

ANEMIC HYPOXIA

- Anemic hypoxia is the condition characterized by the inability of blood to carry enough amount of oxygen. Oxygen availability is normal. But the blood is not able to take up sufficient amount of oxygen due to anemic condition.

CAUSES

- Any condition that causes anemia can cause anemic hypoxia. It is caused by the following conditions:

  i. Decreased number of RBCs

  ii. Decreased hemoglobin content in the blood

  iii. Formation of altered hemoglobin

  iv. Combination of hemoglobin with gases other than oxygen and carbon dioxide.

STAGNANT/ISCHEMIC HYPOXIA

Stagnant hypoxia occurs mainly due to reduction in velocity of blood flow.

CAUSES:

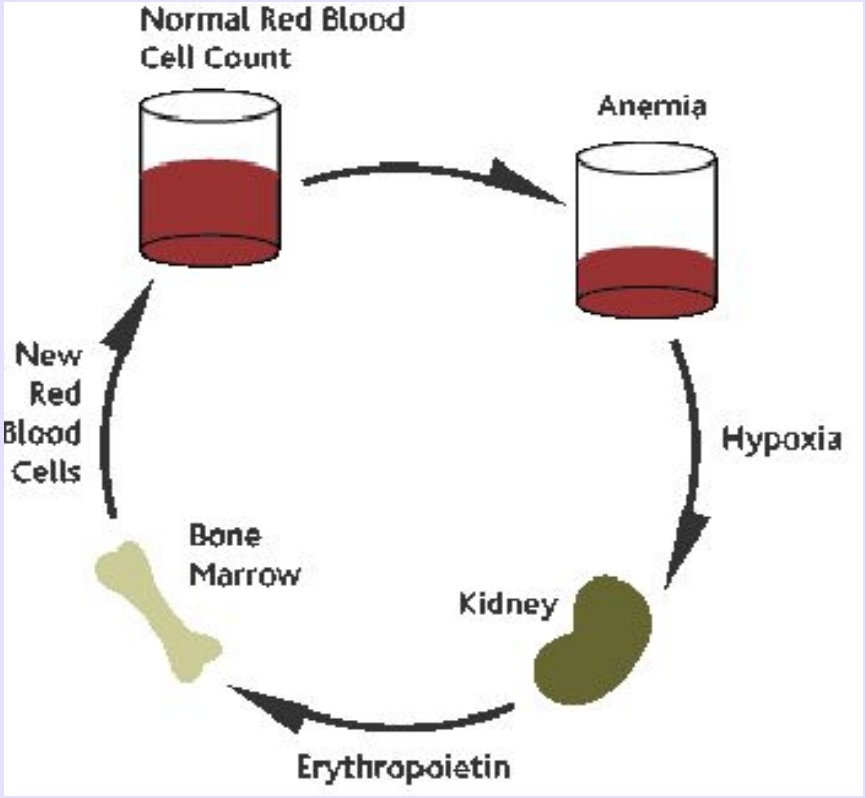
- Decreased cardiac output / sluggish blood flow due heart failure,hemorrhage,circulatory shock and venous obstruction.
- Blood remains in tissues for long time, so tissue extracts increased oxygen from blood, more arteriovenous difference of oxygen concentration.

HISTOTOXIC HYPOXIA

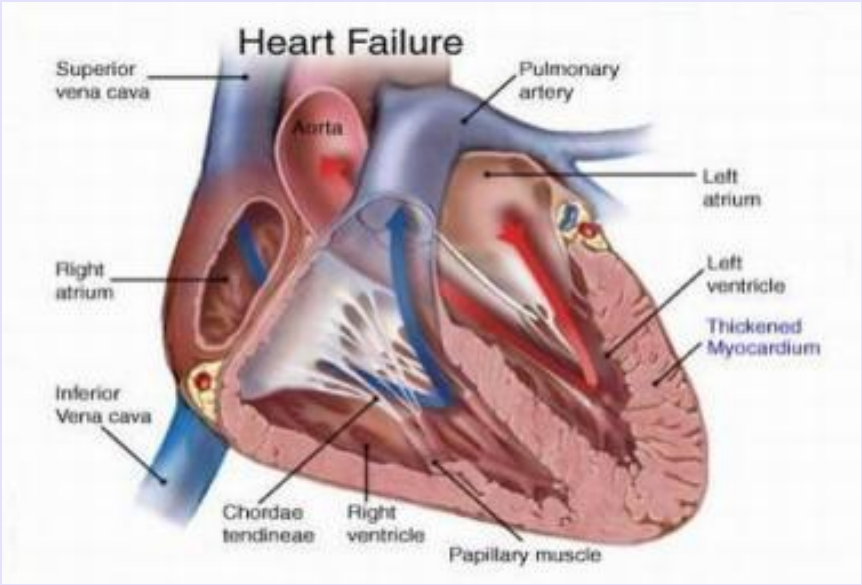
- **DEFINITION**: Inability of the tissue to utilize oxygen inspite of normal arterial PO2 and oxygen carrying capacity.
- **CAUSES:**
- Histotoxic hypoxia occurs due to cyanide or sulfide poisoning. These poisonous substances destroy the cellular oxidative enzymes and there is a complete paralysis of cytochrome oxidase system. So, even if oxygen is supplied, the tissues are not in a position to utilize it..
- Narcotic overdose (it inactivates the enzyme dehydrogenase which causes inhibition of tissue oxygenation)
- Beri-beri (it is deficiency of thymine co-enzymes which is required for many oxidative reaction.)

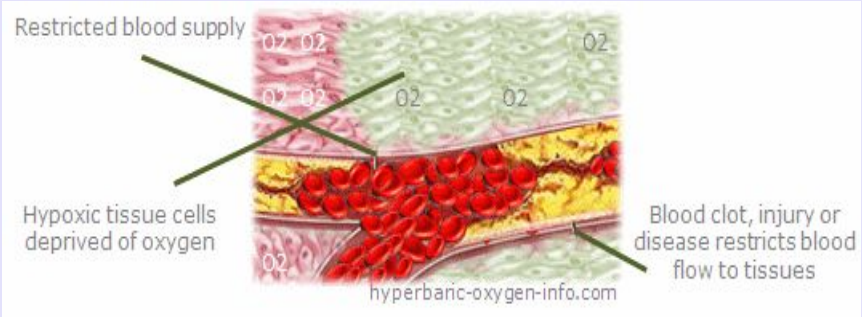
CAUSES / TYPES OF HYPOXIA

ANEMIC HYPOXIA

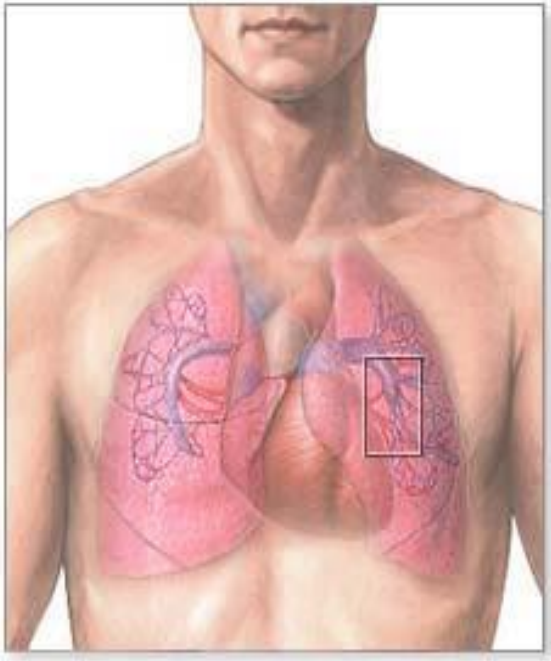


CIRCULATORY HYPOXIA



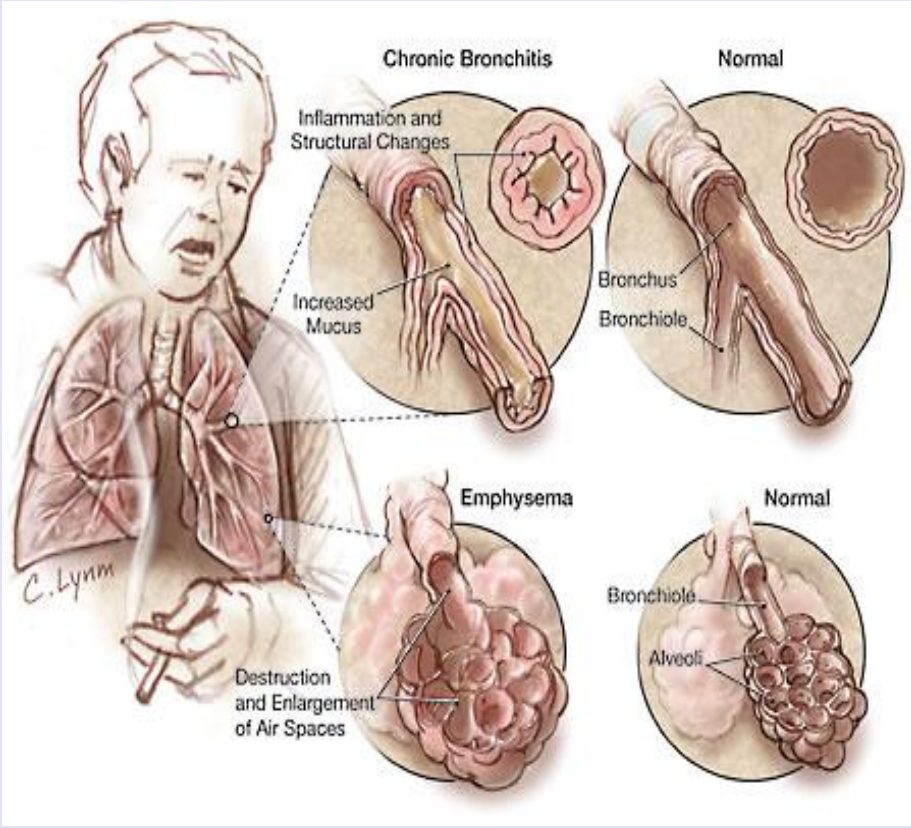


RESPIRATORY HYPOXIA

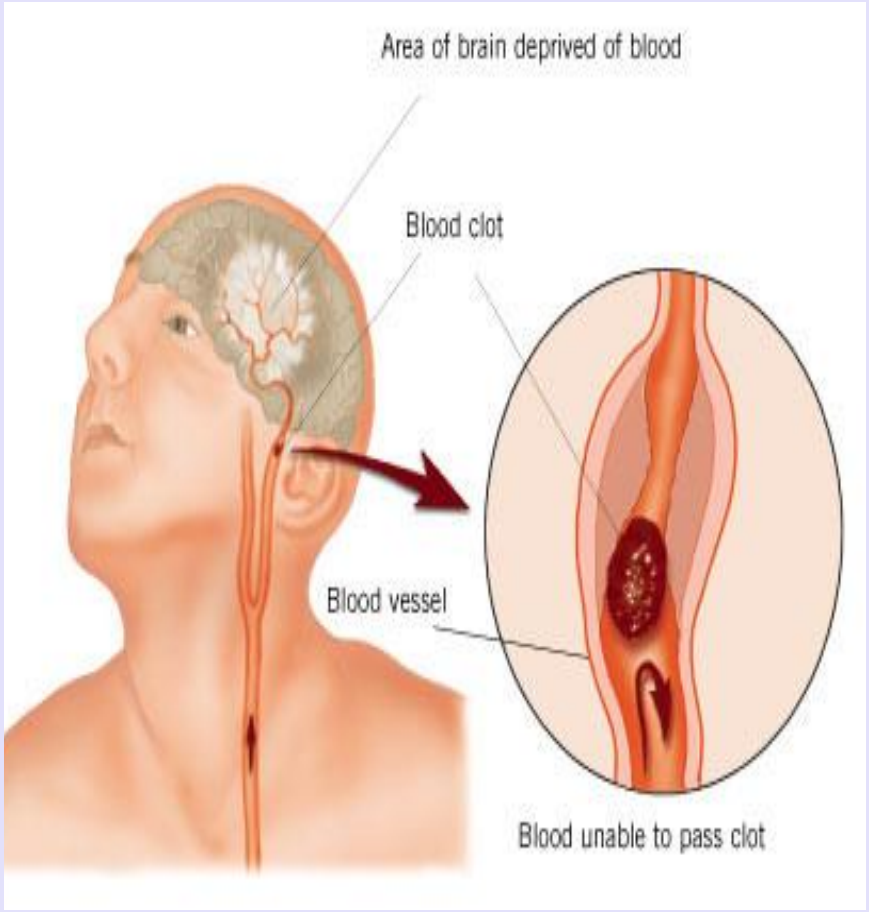




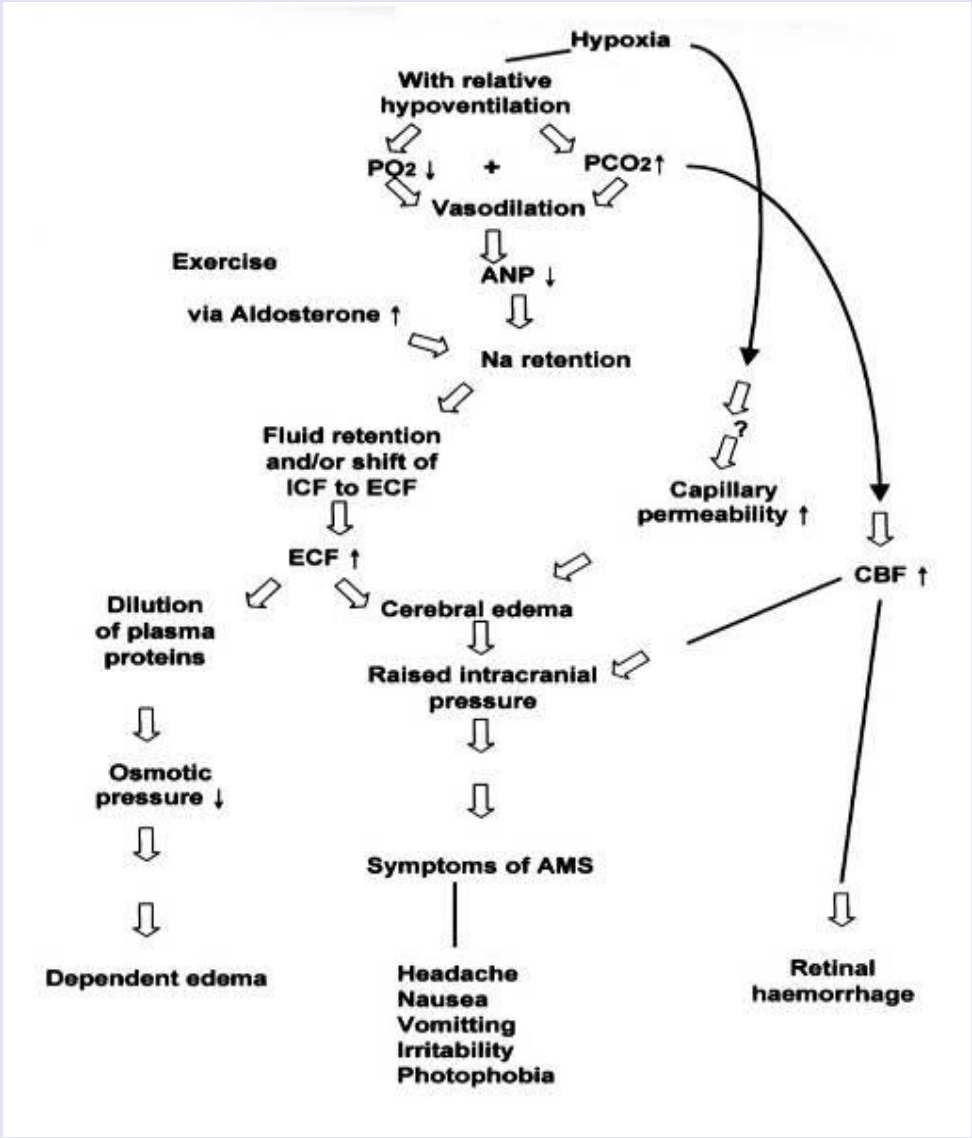
Embolus lodged in left pulmonary artery



CEREBRAL HYPOXIA



HYPOXIA DUE TO HIGH ALTITUDE



HIGH ALTITUDE HYPOXIC EFFECTS

- **Specimen of handwriting at specific altitudes and different percentages of O_2**

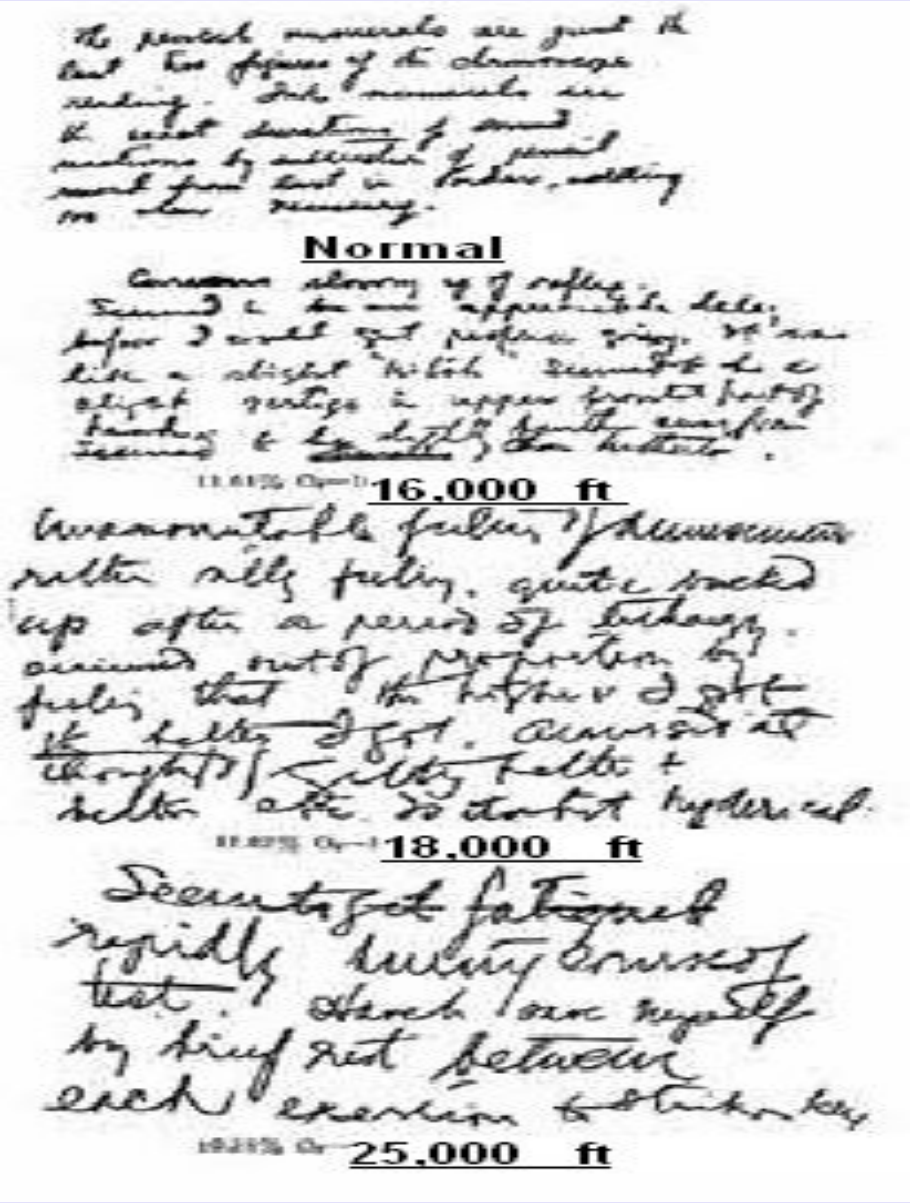


TABLE 5.6 Causes of Hypoxia

Cause	Mechanism	Pa_{O_2}
↓ Cardiac output	↓ Blood flow	—
Hypoxemia	↓ Pa_{O_2} ↓ O_2 saturation of hemoglobin ↓ O_2 content of blood	↓
Anemia	↓ Hemoglobin concentration ↓ O_2 content of blood	—
Carbon monoxide poisoning	↓ O_2 content of blood Left shift of O_2-hemoglobin curve	—
Cyanide poisoning	↓ O_2 utilization by tissues	—

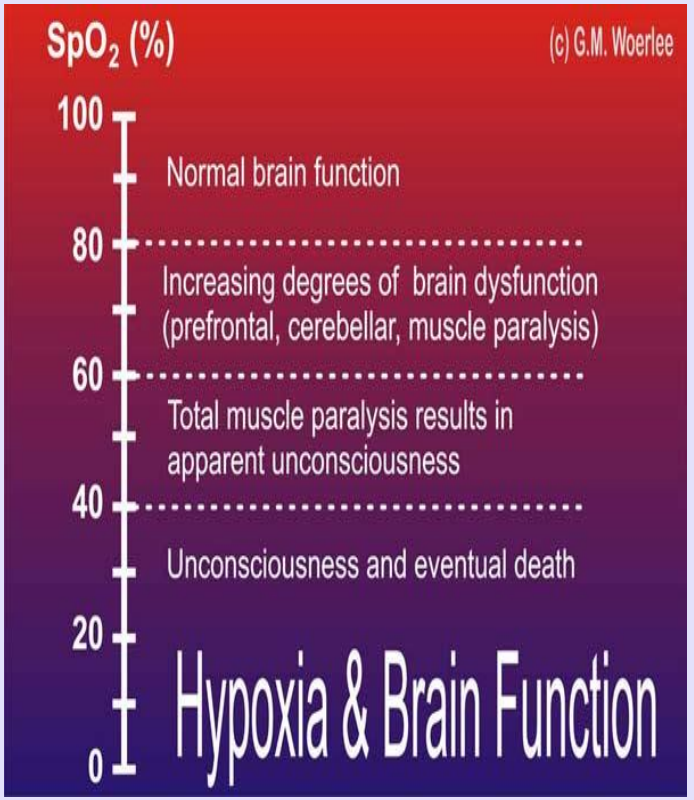
CYANOSIS

- Bluish color of the skin and mucous membranes resulting from an increased quantity of deoxygenated hemoglobin in the skin blood vessels, especially in the capillaries.
- Lips, nail beds, ears, and malar eminences
- 3.4~5.0 g/dl of reduced hemoglobin in the systemic circulation
- You can't see cyanosis until the O2 saturation is in the mid-80**% or less**



EFFECTS OF HYPOXIA

- Changes in the central nervous system,
- Impaired judgment, motor incoordination
- Fatigue, drowsiness, apathy, inattentiveness, delayed reaction time, and reduced work capacity
- Death usually results from respiratory failure (Brainstem hypoxia)



COMPLICATIONS

HYPOXIC-ISCHEMIC ENCEPHALOPATHY

- Impaired judgment, inattentiveness, motor incoordination, and, at times, euphoria
- Circulatory arrest --> consciousness is lost within seconds
- Circulation is restored within 3 to 5 min --> full recovery may occur (eg. Neonatal asphyxia)
- Hypoxia-ischemia lasts beyond 3 to 5 min --> some degree of permanent cerebral damage
- Mild hypothermia

SURVIVAL TIME OF DIFFERENT NERVE TISSUE WITHOUT OXYGEN

TABLE 10

SURVIVAL TIME OF DIFFERENT NERVE TISSUES COMPLETELY DEPRIVED OF BLOOD*

Tissue	Survival Time (Minutes)
Cerebrum, small pyramidal cells	8
Cerebellum, Purkinje's cells	13
Medullary centers	20–30
Spinal cord	45–60
Sympathetic ganglia	60
Myenteric plexus	180

OXYGEN THERAPY IN DIFFERENT TYPES OF HYPOXIA

- In **atmospheric hypoxia**, O2 therapy can completely correct the depressed O2 level in the inspired gases and, therefore, provide 100 percent effective therapy**.**
- In **hypoventilation hypoxia,** a person breathing 100 percent O2 can move five times as much O2 into the alveoli with each breath as when breathing normal air. Therefore, here again O2 therapy can be extremely beneficial.
- In hypoxia caused by **anemia, abnormal hemoglobin transport of O2, circulatory deficiency**, O2 therapy is of much less value because normal O2 is already available in the alveoli.

Cont.

- In hypoxia caused by **inadequate tissue use of O2**, there is abnormality neither of O2 pickup by the lungs nor of transport to the tissues. Instead, the tissue metabolic enzyme system is simply incapable of using the O2 that is delivered. Therefore, O2 therapy provides no measurable benefit.

- https://youtu.be/JcGKDDvk5AQ?si=iVL8VD3RDZ0V9bNZ

THANK YOU